

Mohamed Kechiche

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EDUCATION

Technical University of Darmstadt

Bachelor of Science in Computer Science

- **Coursework:** Algorithms & Data Structures, OOP, Operating Systems, Databases, Concurrent Programming, SWE

Darmstadt, Germany

Oct. 2024 – Sep. 2027

EXPERIENCE

Software Engineer (Working Student), Tech Office

GFT Technologies

- Ship features for global insurance client in TypeScript and Spring Boot, hardened by CI/CD pipelines, automated tests, and structured code review.
- Identified manual bottlenecks across the team's workflows and automated them with AI, reducing repetitive work and speeding up delivery
- Engineer an automated Confluence-to-Git sync service: a REST delta pipeline mirrors the full specification tree into a versioned repo as Markdown every 15 minutes, then an AI agent drafts specs per Jira ticket onto a review branch.

April 2026 – Present

Frankfurt, Germany

Teaching Assistant, Algorithms & Data Structures

Technical University of Darmstadt

- Lead weekly sessions for 35+ students on core algorithms, data structures, and complexity analysis.
- Design and grade 20+ weekly assignments with individual feedback, lifting average scores 30% over the term.
- Run office hours debugging student Java and Python code, resolving 15+ issues per week.

April 2026 – Present

Darmstadt, Germany

Web Developer Intern

GOMYCODE

- Built responsive React features for an e-learning platform serving 1,000+ users; ran cross-browser QA passes.
- Cut production page load times 25% through targeted code refactoring and lazy loading.

June 2023 – Aug. 2023

Sousse, Tunisia

PROJECTS

ARIVA — Cross-Border Document Automation | *Java, Spring Boot, React, FastAPI, Docker*

Feb. 2026

- Scoped requirements over a 3-week build in working sessions with the government caseworkers who would use the tool daily, turning their manual process into a technical design and stack choice.
- Delivered a 3-service system (React client, Spring Boot API, FastAPI pipeline, orchestrated via Docker Compose) that cut case processing from 15–20 minutes to under 90 seconds per document, a 90% reduction.
- Kept sensitive data on-premise behind a pluggable vision client: production runs a locally hosted Qwen3-VL 4B, with a one-flag OpenAI mode used in demos to show the accuracy unlocked by state-provisioned GPU hardware.
- Hardened output with schema-locked generation, confidence scoring, a human review queue for uncertain fields, and an audit record on every state change.

LawBuddy — Personalized Law Impact Tracker (RAG) | *Python, React, Supabase, OpenAI*

March 2026

- Explains new legislation in plain language and shows exactly how it affects each user, including a B2B mode that keeps small businesses compliant as the law changes, replacing hours of manual reading with one query.
- Built a retrieval-augmented generation (RAG) pipeline over a scraped, indexed law database: questions are embedded, matched to the most relevant statutes, and answered only from retrieved text, so every answer cites its source and the model cannot invent law.
- Architected the Supabase backend with real-time auth and a modular React front end, covering retrieval and personalization logic with a Vitest suite.

Lucid Analytics | *React, TypeScript, Supabase, OpenAI* | “Best Idea” & “Best Tech” · Futury Hackathon

April 2026

- Turns a company's raw spreadsheets into a plain-language ranking of where the business is losing money.
- Built-in simulator lets users test a change and see its profit impact before committing; won 1,000 EUR in prizes.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, TypeScript, JavaScript, SQL (PostgreSQL), HTML/CSS, Bash

Frameworks: Spring Boot, React, Node.js, FastAPI, Flask, Angular, Express, TailwindCSS, Pydantic

Testing & CI/CD: JUnit, Vitest, Playwright, pytest, Git, GitHub Actions, Jenkins, Maven

Cloud & Tools: Docker, Kubernetes, Google Cloud Platform, AWS, Linux, PostgreSQL, Qdrant, REST APIs

Data & AI: pandas, NumPy, Matplotlib, OpenCV, RAG pipelines, vector search, embeddings, LLM APIs

Concepts: Data structures & algorithms, OOP, design patterns, distributed systems, microservices, agile/Scrum

Languages (spoken): English (fluent), German (Fluent), French (fluent), Arabic (native)